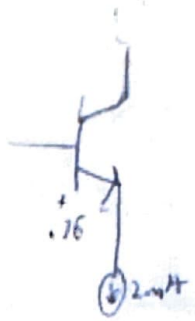
Dylan Trembley

1.



$.002 = I_S e^{\frac{.76}{.025}}$

$I_S = 1.25 \times 10^{-16}$

at .7

$I_1 = 1.25 \times 10^{-16} e^{\frac{.7}{.025}}$

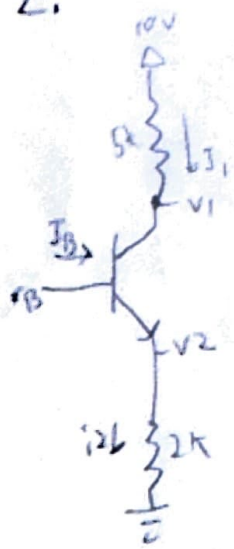
$I_1 = 181.4\ mA$

$4 \times 10^{-6} = 1.25 \times 10^{-16} e^{\frac{V_B}{.025}}$

$24.185 = \frac{V_B}{.025}$

$V_B = .605V$

2.



$V_{BEon} = .7V$ $B = 100$

$V_{CEsat} = .2V$

$V_T = 25mV$

Test for $V_B = 0, 3, 5$

$V_B = 0$ assume active

$0 + .7 + I_E 2k = 0$

$I_E = -mm$ not possible

If B in cutoff everything = 0 for $V_B = 0$

The question asks for V3 and I3 but they don't exist on the circuit.

$V_B = 3$ assume active

$-3 + .7 + I_E 2k = 0$

$3 = .7 + I_E 2k$

$\frac{2.3}{2k} = I_E = 1.15mA = I_2$

$V_E = 2.3V = V_2$ $V_1 = 4.307V = V1$

$I1 = 1.1386mA$ $V_1 > V_B > V_E$ for $V_B = 3$

so our assumption is correct

$1.15mA\left(\frac{100}{101}\right) = 1.1386mA$

$\frac{10 - V1}{5k} = 1.1386$

$V1 = 4.307V$

$V_B = 5$ assume active

$-5 + .7 + I_E 2k = 0$

$I_E = 2.15mA$

$V_E = 4.3V$

$2.15\left(\frac{100}{101}\right) = 2.129mA$

$\frac{10 - V1}{5k} = 2.129mA$

$V1 = -.64$

$V_C < V_B$ so we in saturation

$V_E < V_B$

$V_C = V_E + .2$

$V_1 = 4.3V$

$\frac{10 - 4.5}{5k} = I_1 = 1.1$ mA

$V1 = 4.5V$

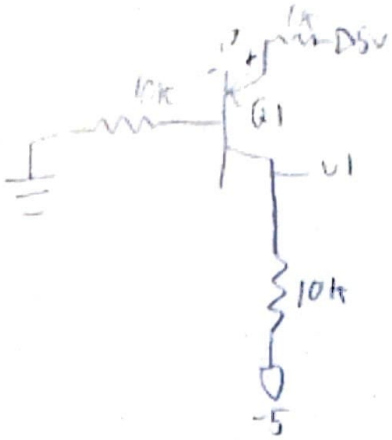
$V2 = 4.3V$

$I2 = 2.15mA$

$I_1 = 1.1mA$

for $V_B = 5$

3.



$V_{EB} = .7V$ $V_T = 25mV$

$V_{E(sat)} = .2V$

$I_B(B+1) = I_E$

Assume Active

$38.74\times10^{-6}(101) = 3.913mA$

$I_C = \frac{100}{101}(3.913\times10^{-3}) = 3.874mA$

$-5 + I_E(1k) + .7 + I_B(10k) = 0$

$-5 + 101I_B(1k) + .7 + 10I_B(1k) = 0$

$111I_B(1k) = 4.3$

$I_B = \frac{4.3}{111000}$

$I_B = 38.74\mu A$

$\frac{V_B - 0}{10k} = 38.74\mu A$

$V_B = .387V$

$\frac{5 - V_E}{1k} = 3.913mA$

$V_E = 1.087V$

$V_E > V_B$

$V_C > V_B$ so it's saturated

$\frac{V_C + 5}{10k} = 3.874mA$

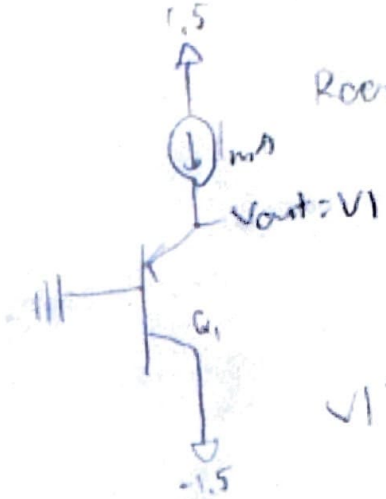
$V_C = 33.739V$

$1.087 - .2 = V_C = .887$

$\frac{.887 - (-5)}{10k} = .589mA$

$\boxed{V_C = .887V \text{ and in saturation mode}}$

4.



Room temp = 20° C = 293K

$$V_T = \frac{kT}{q} = \frac{1.38\times10^{-23}\,T}{1.6\times10^{-19}}$$

V1 = Vout

$$V_1 = V_T \ln\left(\frac{1\times10^{-3}}{1\times10^{-5}}\right)$$

$$\frac{293\cdot(1.38\times10^{-23})}{1.6\times10^{-19}} =$$

$V_1 = V_T(27.63)$ $\quad V_T = .025$

a. V1 = .691V

b. $V_1 = V_T \ln\left(\frac{1\times10^{-3}}{1\times10^{-19}}\right)$

$V_1 = V_T(27.63)$

$V_1 = .812V$

$$V_T = \frac{343\cdot(1.38\times10^{-23})}{1.6\times10^{-19}} =$$

$V_T = .0296$

$$V_T = \frac{243\cdot(1.38\times10^{-23})}{1.6\times10^{-19}}$$

$V_T = .0209$ ✓

c.

$V_1 = V_T \ln\left(\frac{1\times10^{-3}}{1\times10^{-5}}\right)$

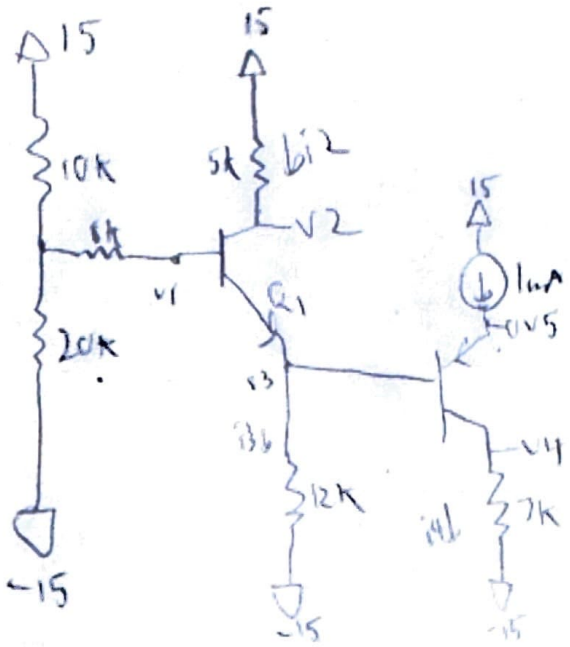
$V_1 = V_T(27.63)$

$V_1 = .579V$

3. Sorry for bad work flow

$V_{BE/EB} = .7V$ $V_{CEsat} = .2V$

$\beta = 100$ $V_T = 25mV$

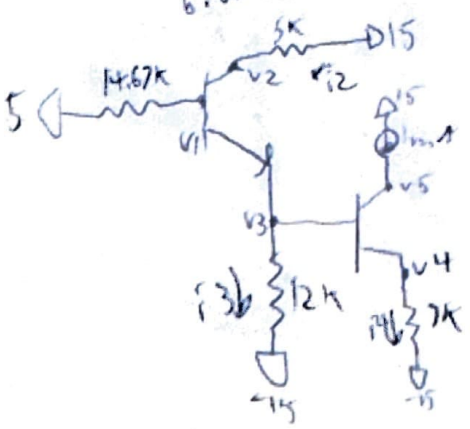


$15 - (-15) = 30$

$30\left(\frac{20k}{30k}\right) = 20 - 15 = 5V$

$10k \| 20k = 6.6k$

$6.67k + 8 = 14.67k$



start assuming active

$I_{E2} = 1mA$

$I_{B2} = \frac{I_{E2}}{101}$

$I_{B2} = 9.9\mu A$

$1mA - 9.9\mu A = .99mA = I_4$

$\frac{V_4 + 15}{7k} = .99mA$

$V_4 = -8.069V$

$4.769 - .7 = 4.069V = V_3$

$4.769 = V_5$

$\frac{4.069 + 15}{12k} = I_3 = 1.589mA = I_1$

$1.589 - .99mA = 1.579mA = I_2$

$I_{C1} = \frac{100}{101} I_{E1} =$

$\frac{15 - V_2}{5k} = 1.563mA$

$V_2 = 7.133V$

b/c $V_{C1} = V_2 > V_1 = V_{B1} > V_3 = V_E$ for the NPN, the NPN is active and b/c $V_{C2} = V_4 < V_3 = V_{E1} = V_{B2} < V_{E2} = V_5$

$V_2 = 7.133$, $V_1 = V_{B1} = 4.769$, $V_3 = V_E = 4.069$

$V_4 = -8.069$, $V_3 = 4.069$, $V_5 = 4.769$

for the PNP, the PNP is also active

$V1 = 4.769V$	$I1 = 15.7\mu A$
$V2 = 7.185V$	$I2 = 1.563mA$
$V3 = 4.069V$	$I3 = 1.589mA$
$V4 = -8.069V$	$I4 = .99mA$
$V5 = 4.769V$	

$-5 + I_{B1}(14.67k) + .7 + I_{E1}(12k) - 15 = 0$

$-5 + I_{B1}(14.67k) + .7 + 101 I_{B1}(12k) - 15 =$

$I_{B1}(14.67k + 101(12k)) = 19.3$

$I_{B1} = 15.7\mu A$

$\frac{5 - V1}{14.67k} = 15.7\mu A$

$V1 = V_{B1} = 4.769V$